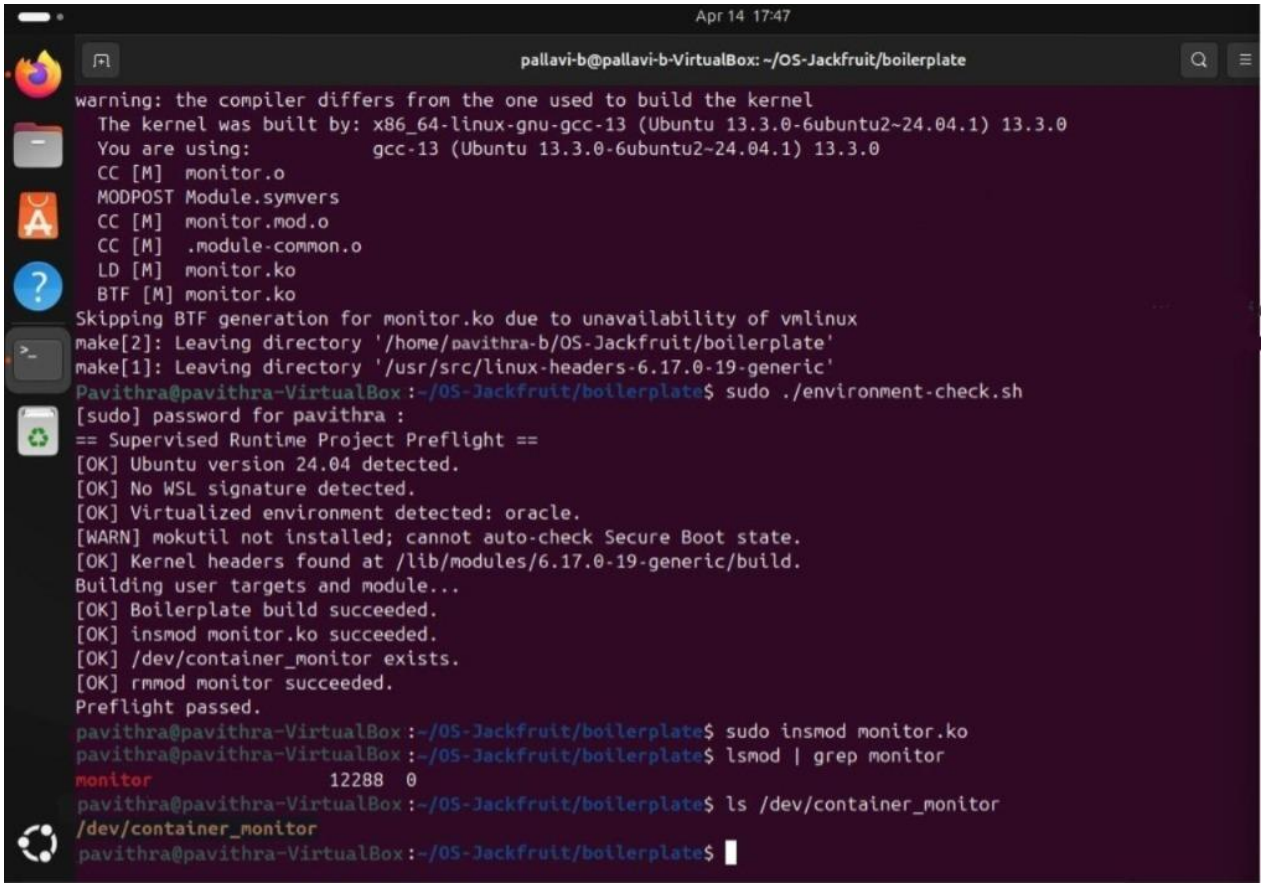
# OS-JACKFRUIT

## Multi-Container Runtime

**Team Mem1:K R Tanishkaa SRN:PES1UG24AM126 SEC: C**

**Team Mem2:Pavithra B SRN:PES1UG25AM807 SEC: C**

Screenshots:




TASK 1:

```
pavithra@pavithra-VirtualBox :~/OS-Jackfruit/boilerplate$ sudo ./engine start c1 rootfs "cpu_hog 20"
[sudo] password for pallavi-b:
ok
pavithra@pavithra-VirtualBox :~/OS-Jackfruit/boilerplate$ sudo ./engine start c2 rootfs "cpu_hog 20"
ok
pavithra@pavithra-VirtualBox :~/OS-Jackfruit/boilerplate$ sudo ./engine ps
c2                pid=9340   state=exited   soft= 40MiB hard= 64MiB code=127 sig=0   start=18:00:26
c1                pid=9289   state=exited   soft= 40MiB hard= 64MiB code=127 sig=0   start=17:59:40
```

```
pavithra@pavithra-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine start c4 rootfs "/cpu_hog 100"
ok
pavithra@pavithra-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
c4                pid=15111  state=running  soft= 40MiB hard= 64MiB code=0   sig=0   start=18:17:11
c3                pid=11641  state=exited   soft= 40MiB hard= 64MiB code=0   sig=0   start=18:15:28
c2                pid=10204  state=exited   soft= 40MiB hard= 64MiB code=0   sig=0   start=18:08:35
c1                pid=10180  state=exited   soft= 40MiB hard= 64MiB code=0   sig=0   start=18:08:25
c2                pid=9340   state=exited   soft= 40MiB hard= 64MiB code=127 sig=0   start=18:00:26
c1
pavithra@pavithra-VirtualBox:~/OS-Jackfruit/boilerplate$
```

TASK 2:

```
pavithra@pavithra-VirtualBox :~/OS-Jackfruit/boilerplate$ sudo ./engine ps
c2                pid=10204  state=exited   soft= 40MiB hard= 64MiB code=0   sig=0   star
c1                pid=10180  state=exited   soft= 40MiB hard= 64MiB code=0   sig=0   star
c2                pid=9340   state=exited   soft= 40MiB hard= 64MiB code=127 sig=0   star
c1                pid=9289   state=exited   soft= 40MiB hard= 64MiB code=127 sig=0   star

pavithra@pavithra-VirtualBox :~/OS-Jackfruit/boilerplate$ ls   rootfs/
alpine-minirootfs-3.23.3-x86_64.tar.gz  cpu_hog  etc   io_pulse  media       mnt  proc
bin                                     dev      home  lib       memory_hog  opt  root
pavithra@pavithra-VirtualBox :~/OS-Jackfruit/boilerplate$ sudo ./engine logs c1
/bin/sh: cpu_hog: not found
cpu_hog alive elapsed=1 accumulator=8488015175725468995
cpu_hog alive elapsed=2 accumulator=6033449358966712512
cpu_hog alive elapsed=3 accumulator=4339841198134536119
cpu_hog alive elapsed=4 accumulator=7292795584196533636
cpu_hog alive elapsed=5 accumulator=8615908445189209954
cpu_hog alive elapsed=6 accumulator=2397641973987662503
cpu_hog alive elapsed=7 accumulator=17675932111116908633
cpu_hog alive elapsed=8 accumulator=12075446813871187839
cpu_hog alive elapsed=9 accumulator=11390401373559937977
cpu_hog alive elapsed=10 accumulator=6427755514765479242
cpu_hog alive elapsed=11 accumulator=5547866628412097498
cpu_hog alive elapsed=12 accumulator=13225349365678508617
cpu_hog alive elapsed=13 accumulator=14462013591419234240
cpu_hog done duration=20 accumulator=6801000933385675702
pavithra@pavithra-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine stop c1
container not found or not running
pavithra@pavithra-VirtualBox:~/OS-Jackfruit/boilerplate$
```

TASK 3:

```
pavithra@pavithra-VirtualBox :~/OS-Jackfruit/boilerplate$ sudo ./engine run c3 rootfs "/cpu_hog 5"
[run] waiting for container to finish…
state=exited code=0 signal=0
pavithra@pavithra@VirtualBox :~/OS-Jackfruit/boilerplate$ cat logs/c3.log
cpu_hog alive elapsed=1 accumulator=6714212733155315941
cpu_hog alive elapsed=2 accumulator=6618240562122467652
cpu_hog alive elapsed=3 accumulator=1623912300560331359 6
cpu_hog alive elapsed=4 accumulator=9514322129083419 71
cpu_hog alive elapsed=5 accumulator=1810959324513087 1426
cpu_hog alive elapsed=6 accumulator=686108703922880681
cpu_hog alive elapsed=7 accumulator=3401053677491740527
cpu_hog alive elapsed=8 accumulator=1223303865565896 6537
cpu_hog alive elapsed=9 accumulator=2041230254743977408
cpu_hog alive elapsed=13 accumulator=7170637677605683577
cpu_hog alive elapsed=14 accumulator=16393574937059400447
cpu_hog alive elapsed=15 accumulator=11827388193847270057
cpu_hog alive elapsed=16 accumulator=14377626064443733320
cpu_hog alive elapsed=17 accumulator=13522037886052027350
cpu_hog alive elapsed=18 accumulator=7607683809324154061
cpu_hog alive elapsed=19 accumulator=12651013183571407044
cpu_hog alive elapsed=20 accumulator=2518993587427591787
cpu_hog alive elapsed=21 accumulator=4434451414469635023
```

```
Apr 14 18:21
pallavi-b@pallavi-b-VirtualBox: ~/OS-Jackfruit/boilerplate
pallavi-b@pallavi-b-VirtualBox: ~/OS-Jackfruit/boilerplate   pallavi-b@pallavi-b-VirtualBox: ~/OS-Jackfruit/boilerplate
cpu_hog alive elapsed=25 accumulator=6534811994822408123
cpu_hog alive elapsed=26 accumulator=6760259344068015841
cpu_hog alive elapsed=30 accumulator=1147220970640167515
cpu_hog alive elapsed=31 accumulator=5364839943822199549
cpu_hog alive elapsed=32 accumulator=7264092236270666038
cpu_hog alive elapsed=33 accumulator=4499155384538955452
cpu_hog alive elapsed=34 accumulator=14639179011983481384
cpu_hog alive elapsed=35 accumulator=12865104147413858319
cpu_hog alive elapsed=36 accumulator=12712876961592311047
cpu_hog alive elapsed=37 accumulator=11285442476416715 81
cpu_hog alive elapsed=38 accumulator=3868583972042116 78
cpu_hog alive elapsed=39 accumulator=6015178973589781073
cpu_hog alive elapsed=54 accumulator=15006476381075981262
cpu_hog alive elapsed=55 accumulator=17918464280488886155
cpu_hog alive elapsed=56 accumulator=632456040251411127
cpu_hog alive elapsed=57 accumulator=7384659404389094684
cpu_hog alive elapsed=58 accumulator=1096823671401906327
cpu_hog alive elapsed=59 accumulator=9557075351983169063
cpu_hog alive elapsed=60 accumulator=411175023313787733
cpu_hog alive elapsed=61 accumulator=7619184934965498454
cpu_hog alive elapsed=62 accumulator=1502981503678653814
cpu_hog alive elapsed=63 accumulator=1768525071273811290
cpu_hog alive elapsed=64 accumulator=5842404482700947076
cpu_hog alive elapsed=65 accumulator=4295718307275054603
cpu_hog alive elapsed=79 accumulator=13487021162472607042
cpu_hog alive elapsed=80 accumulator=10544385020702347777
cpu_hog alive elapsed=81 accumulator=1704060486756370775
cpu_hog alive elapsed=82 accumulator=13684449482333565960
cpu_hog alive elapsed=83 accumulator=3608759285278802025
cpu_hog alive elapsed=84 accumulator=11782575117011700673
cpu_hog alive elapsed=85 accumulator=14185679886525056786
```

# TASK 4:

```
pavithta@pavithra-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine run m1 rootfs "/memory_hog 8 500"
[run] waiting for container to finish…
state=killed code=0 signal=9
pavithra@pavithra-VirtualBox:~/OS-Jackfruit/boilerplate$ dmesg | tail
dmesg: read kernel buffer failed: Operation not permitted
pavithra@pavithra-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo dmesg | tail
[ 4258.085869] [container_monitor] pid=16902 (container=c3) gone, removing entry
[ 4292.143385] workqueue: vmstat_update hogged CPU for >10000us 7 times, consider switching to WQ_UNBOUND
[ 4380.408484] [container_monitor] Registered container=m1 pid=17838 soft=41943040 hard=67108864
[ 4382.867403] [container_monitor] SOFT LIMIT container=m1 pid=17838 rss=42489984 limit=41943040
[ 4384.915134] [container_monitor] HARD LIMIT container=m1 pid=17838 rss=75964416 limit=67108864 - SIGKILL sent
[ 4393.101766] workqueue: delayed_vfree_work hogged CPU for >10000us 7 times, consider switching to WQ_UNBOUND
[ 4434.611571] audit: type=1400 audit(1776171183.890:414): apparmor="DENIED" operation="open" class="file" profile="snap
.firefox.firefox" name="/proc/pressure/memory" pid=4141 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=100
0 ouid=0
[ 4444.967648] audit: type=1400 audit(1776171194.251:415): apparmor="DENIED" operation="open" class="file" profile="snap
.firefox.firefox" name="/proc/pressure/memory" pid=4141 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=100
0 ouid=0
[ 4482.619376] audit: type=1400 audit(1776171231.904:416): apparmor="DENIED" operation="open" class="file" profile="snap
.firefox.firefox" name="/proc/pressure/memory" pid=4141 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=100
0 ouid=0
[ 4487.622521] audit: type=1400 audit(1776171236.907:417): apparmor="DENIED" operation="open" class="file" profile="snap
.firefox.firefox" name="/proc/pressure/memory" pid=4141 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=100
0 ouid=0
pavithra@pavithra-VirtualBox:~/OS-Jackfruit/boilerplate$
```

# TASK 5:

```
pavithra@pavithra-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine start c8 rootfs "/cpu_hog 30" --nice 10
sudo ./engine start c9 rootfs "/cpu_hog 30" --nice 10
sudo ./engine start c10 rootfs "/cpu_hog 30" --nice -5
ok
ok
ok
pavithra@pavithra-VirtualBox:~/OS-Jackfruit/boilerplate$ top

top - 18:32:59 up  1:23,  1 user,  load average: 1.05, 0.98, 1.75
Tasks: 233 total,   6 running, 226 sleeping,   1 stopped,   0 zombie
%Cpu(s): 28.2 us, 25.6 sy, 35.9 ni,  0.0 id,  0.0 wa,  0.0 hi, 10.3 si,  0.0 st
MiB Mem :   3914.8 total,    384.7 free,   2233.9 used,   1395.8 buff/cache
MiB Swap:   3916.0 total,   3550.8 free,    365.2 used.   1680.9 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
  18547 root      15  -5    1012    728    676 R  94.9   0.0   0:06.67 cpu_hog
  18534 root      30  10    1012    732    676 R  71.5   0.0   0:05.29 cpu_hog
  18524 root      30  10    1012    724    676 R  24.9   0.0   0:01.42 cpu_hog
   5012 pavithra+ 20   0 3523016 769368 118476 R   3.7  19.2  24:44.75 Isolated Web Co
   2623 pavithra+ 20   0 3905632 169764  89108 R   2.0   4.2   6:43.44 gnome-shell
    244 root      19  -1   51460  17396  16624 S   0.8   0.4   0:15.23 systemd-journal
  18554 pavithra+ 20   0   14520   6036   3820 R   0.8   0.2   0:00.08 top
```

```
MiB Swap:   3916.0 total,   3550.8 free,    365.2 used.   1680.9 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
  18547 root      15  -5    1012    728    676 R  94.9   0.0   0:06.67 cpu_hog
  18534 root      30  10    1012    732    676 R  71.5   0.0   0:05.29 cpu_hog
  18524 root      30  10    1012    724    676 R  24.9   0.0   0:01.42 cpu_hog
   5012 pavithra+ 20   0 3523016 769368 118476 R   3.7  19.2  24:44.75 Isolated Web Co
   2623 pavithra+ 20   0 3905632 169764  89108 R   2.0   4.2   6:43.44 gnome-shell
    244 root      19  -1   51460  17396  16624 S   0.8   0.4   0:15.23 systemd-journal
  18554 pavithra+ 20   0   14520   6036   3820 R   0.8   0.2   0:00.08 top
   6914 pavithra+ 20   0  701452  54116  41236 S   0.6   1.3   0:40.32 gnome-terminal-
     15 root      20   0       0      0      0 I   0.3   0.0   0:21.30 rcu_preempt
   4141 pavithra+ 20   0   11.9g 520116 201432 S   0.3  13.0  12:58.29 firefox
   4461 pavithra+ 20   0 2697204 151192 101852 S   0.3   3.8   0:36.20 Privileged Cont
   5683 pavithra+ 20   0 2644756  93024  66040 S   0.3   2.3   0:13.96 Isolated Servic
   9599 pavithra+ 20   0 1457544 173664 108536 S   0.3   4.3   0:14.08 nautilus
  10833 root      20   0       0      0      0 I   0.3   0.0   0:03.91 kworker/u9:6-events_freezable_pwr_efficient
      1 root      20   0   23468  15016   9816 S   0.0   0.4   0:16.00 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.35 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.00 kworker/u8:0-ipv6_addrconf
[2]+  Stopped                 top
```

# TASK 6:

```
./var/log/
pavithra@pavithra-VirtualBox:~/OS-Jackfruit/boilerplate/rootfs$ ls
alpine-minirootfs-3.23.3-x86_64.tar.gz  dev  home  media  opt  root  sbin  sys  usr
bin                                     etc  lib   mnt    proc run   srv   tmp  var
pavithra@pavithra-VirtualBox:~/OS-Jackfruit/boilerplate/rootfs$ cd ..
pavithra@pavithra-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo cp memory_hog cpu_hog io_pulse rootfs/
pavithra@pavithra-VirtualBox:~/OS-Jackfruit/boilerplate$ ls rootfs/
alpine-minirootfs-3.23.3-x86_64.tar.gz  cpu_hog  etc   io_pulse  media       mnt  proc  run   srv  tmp  var
bin                                     dev      home  lib       memory_hog  opt  root  sbin  sys  usr
pavithra@pavithra-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor rootfs
[supervisor] ready (rootfs=rootfs socket=/tmp/mini_runtime.sock)
[supervisor] started 'c1' pid=9289
[supervisor] started 'c2' pid=9340
[supervisor] started 'c1' pid=10180
[supervisor] started 'c2' pid=10204
[supervisor] started 'c3' pid=11641
Container 'c3' already running
Container 'c3' already running
Container 'c3' already running
[supervisor] started 'c4' pid=15111
[supervisor] started 'c3' pid=16902
[supervisor] started 'm1' pid=17838
[supervisor] started 'c4' pid=18172
[supervisor] started 'c5' pid=18216
[supervisor] started 'c8' pid=18524
[supervisor] started 'c9' pid=18534
[supervisor] started 'c10' pid=18547
```

^C[supervisor] shutting down pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$